

Software Test Plan (STP)

Blood Bank Management System

Version: 1.0

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Date: 16-09-2026

Status: Draft

1. Introduction

Purpose: This document defines the test plan for the Blood Bank Management System. It describes what will be tested, the testing approach, environment, responsibilities, risks, traceability, and criteria for starting and completing testing.

Scope: Testing covers registration and authentication, role-based access, donor and patient profiles, blood donation submission and review, blood requests and review, blood availability, inventory updates, donation/request history, administrator functions, expiry tracking, security, data integrity, usability, and performance. Physical blood collection, laboratory testing, transportation, and clinical decision-making are excluded.

References: Blood Bank Management System SRS v1.0 and the Software Test Plan template provided for the Software Engineering mini project.

Definitions: SRS (Software Requirements Specification), STP (Software Test Plan), FR (Functional Requirement), NFR (Non-Functional Requirement), SR (Security Requirement), RTM (Requirements Traceability Matrix), UAT (User Acceptance Testing).

2. Test Items

- User registration and authentication module
- Role-based authorization for Admin, Donor, and Patient
- Donor profile and donation module
- Patient profile and blood-request module
- Blood inventory module
- Donation and request approval module
- Admin dashboard and user-management functions
- Blood expiry tracking
- Database/data-integrity functions
- Web user interface

3. Features to be Tested

Requirement ID	Feature
BB-F-001	User Registration
BB-F-002	User Login
BB-F-003	Role-Based Access
BB-F-004	Donor Profile
BB-F-005	Donation Submission

BB-F-006	Donation Review / Approval or Rejection
BB-F-007	Donation History
BB-F-008	Patient Profile
BB-F-009	Blood Request
BB-F-010	Request Review / Approval or Rejection
BB-F-011	Blood Availability
BB-F-012	Inventory Update
BB-F-013	Donation Stock Update
BB-F-014	Request Stock Deduction
BB-F-015	Request History
BB-F-016	Admin Dashboard
BB-F-017	User Management
BB-F-018	Expiry Tracking
BB-NF-001	Performance
BB-NF-002	Reliability / Data Preservation
BB-NF-003	Password and Authentication Security
BB-NF-004	Usability
BB-NF-005	Maintainability
BB-NF-006	Data Integrity
BB-SR-001	Password Protection
BB-SR-002	Authenticated Access
BB-SR-003	Role-Based Authorization
BB-SR-004	Input Validation
BB-SR-005	HTTPS in network deployment

4. Features Not to be Tested

- Physical blood collection and handling procedures.
- Laboratory screening and medical testing of donated blood.
- Blood transportation and logistics outside the software system.
- Clinical decisions about whether a patient should receive a particular blood product.
- External services or infrastructure not part of the implemented project.

5. Test Approach / Strategy

5.1 Testing Levels

- Unit Testing: Test individual Django functions, views, forms, models, and small components.
- Integration Testing: Test interactions between authentication, donation, request, inventory, and database components.
- System Testing: Test complete user workflows through the web application.
- Acceptance Testing (UAT): Verify that the implemented system satisfies the agreed SRS requirements from the user's perspective.

5.2 Testing Types

- Functional Testing: Verify each required function produces the expected result.
- Regression Testing: Re-run important tests after changes or defect fixes.
- Performance Testing: Check normal response-time behavior against the SRS target.
- Usability Testing: Check navigation, form clarity, validation messages, and ease of use.
- Data Integrity Testing: Verify blood stock and user/request/donation records remain consistent.
- Security Testing: Verify authentication, authorization, password protection, input validation, and secure deployment.

5.3 Entry Criteria

- Stable application build is available.
- Required test environment is installed and working.

- Test users and sample blood-bank data are available.
- Relevant SRS requirements and acceptance criteria are finalized.

5.4 Exit Criteria

- 100% of planned test cases executed.
- All high-priority functional requirements tested.
- No critical defects remain open.
- Major acceptance criteria satisfied.
- Final test results and RTM updated.

5.5 Security Validation

- Verify passwords are protected using Django's password-handling mechanisms.
- Verify protected pages require authentication.
- Verify donors and patients cannot access administrator-only functions.
- Test invalid and unexpected form inputs.
- Verify HTTPS is used when the system is deployed over a network.

6. Test Environment

Category	Planned Environment
Hardware	Standard laptop/desktop computer for development and testing.
Operating System	Development/testing operating system supported by the team.
Software	Python, Django application, configured database, web browser, and project dependencies.
Browser	Modern browser such as Google Chrome or Microsoft Edge.
Testing Tools	Django test framework / Python testing tools, browser developer tools, and manual testing.
Test Data	Dummy donor, patient, administrator, donation, request, blood-group, inventory, and expiry records.

7. Test Schedule

Milestone	Planned Activity
Test case design	Prepare test cases from SRS requirements and acceptance criteria.
Test environment setup	Configure Django application, database, browser, and test data.
Smoke testing	Check main application pages and login flow.
Test execution	Execute functional, integration, security, usability, performance, and data-integrity tests.
Defect retesting	Retest failed cases after corrections.
Regression testing	Run important existing tests after changes.
UAT / Final verification	Verify final system against SRS acceptance criteria.
Test summary	Prepare final results, metrics, unresolved defects, and sign-off information.

8. Test Deliverables

- Software Test Plan (this document)
- Test cases

- Test data
- Test execution records/logs
- Defect reports
- Updated Requirements Traceability Matrix
- Test summary report

9. Roles and Responsibilities

Team Member	Primary Responsibility
Member 1 / QA Lead	Test planning, test items, and requirement coverage review
Member 2 / Test Engineer	Test strategy, security validation, and environment planning
Member 3 / Developer / Test Support	Schedule, deliverables, risks, and assumptions/dependencies
Member 4 / Traceability & Reporting	Traceability, metrics, reporting, suspension/resumption, and approvals

10. Risks and Mitigation

Risk	Mitigation
Unstable build or frequent code changes	Use smoke testing first and maintain a stable test build before full execution.
Insufficient test data	Prepare dummy users, blood groups, donations, requests, and inventory records before testing.
Database/data inconsistency	Use controlled test data and verify inventory changes after relevant transactions.
Testing schedule delay	Prepare test cases during development so execution can start after a stable build.
Defects found late	Run functional and integration tests early and retest fixes promptly.
Team member availability	Keep test cases and results shared among all four members so testing can continue.

11. Assumptions & Dependencies

- The Django application and required database will be available for testing.
- Test accounts for Administrator, Donor, and Patient will be available.
- Sample blood inventory data will be available.
- SRS requirement IDs remain stable during test planning.
- The team will have access to the source code and required test environment.

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable.
- A blocking defect prevents execution of a significant portion of planned tests.
- The build is too unstable for trustworthy results.
- Required test data or database services are unavailable.

Resume testing when:

- Blocking defects are resolved or a stable build is supplied.
- The test environment is restored.
- Required test data and database services are available.

13. Test Case Management & Traceability

Test cases shall be linked to SRS requirement IDs so that every important requirement can be verified. The RTM will be updated with test case IDs and execution status.

SRS Requirement	Test Case(s)	Purpose
BB-F-001	TC-REG-01	Verify valid and invalid user registration.
BB-F-002	TC-AUTH-01	Verify successful and unsuccessful login.
BB-F-003	TC-AUTH-02	Verify role-based access restrictions.
BB-F-005	TC-DON-01	Verify donation submission.
BB-F-006	TC-DON-02	Verify donation approval/rejection.
BB-F-009	TC-REQ-01	Verify blood-request submission.
BB-F-010	TC-REQ-02	Verify request approval/rejection.
BB-F-011	TC-INV-01	Verify blood availability display.
BB-F-013	TC-INV-02	Verify stock increase after approved donation.
BB-F-014	TC-INV-03	Verify stock deduction after approved request.
BB-NF-001	TC-PERF-01	Verify response-time target under normal test load.
BB-NF-006	TC-DATA-01	Verify inventory and required-field data integrity.
BB-SR-003	TC-SEC-03	Verify administrator-only authorization.

14. Test Metrics & Reporting

Metrics to be collected:

- Percentage of planned test cases executed.
- Percentage of test cases passed and failed.
- Number of open, closed, and reopened defects.
- Requirement coverage based on the RTM.
- Number of critical/high-priority defects.
- Defect aging where applicable.

Reports:

- Test execution status during the testing period.
- Defect status and retest results.

- Final Test Summary Report.

15. Approvals

Role	Name	SRN
Member 1 / QA Lead	KHUSH BHARDWAJ	PES2UG24CS226
Member 2 / Test Engineer	KOUSHIK C	PES2UG24CS232
Member 3 / Developer / Test Support	QAMAR AHMED	PES2UG24CS904
Member 4 / Traceability & Reporting	Hemaksh Breja	PES2UG24CS191
Course Coordinator / Instructor	Sheela Devi	-

Division of Work

Team Member	Responsibility in this Test Plan
Member 1	Introduction, Test Items, Features to be Tested, and review
Member 2	Test Approach/Strategy, Security Validation, and Test Environment
Member 3	Test Schedule, Test Deliverables, Risks, and Assumptions/Dependencies
Member 4	Suspension/Resumption, Traceability, Metrics/Reporting, Approvals, and review

All four team members participate in final review, correction of inconsistencies, and preparation of the final submission.